

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

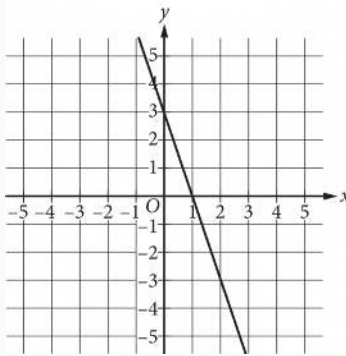
10

Q1

ALGEBRA

For a 3-week period in a town in Illinois, the lowest recorded temperature was 31 degrees Fahrenheit °F and the highest recorded temperature was 67°F. Which inequality is true for any recorded temperature t , in °F, in this town for this 3-week period?

- A. $t \geq 98$
- B. $t \geq 67$
- C. $31 \leq t \leq 67$
- D. $t \leq 31$



What is the equation of the line shown in the xy -plane above?

- A. $y = 3x - 3$
- B. $y = -3x + 3$
- C. $y = \frac{1}{3}x - 3$
- D. $y = -\frac{1}{3}x + 3$

Q3

GRID-IN

ALGEBRA

If $6 + x = 9$, what is the value of $18 + 3x$?

STUDENT-PRODUCED RESPONSE

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Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A contract for a certain service requires a onetime activation cost of \$ 35 and a monthly cost of \$ 23. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

A. $c = \frac{t}{23} + 35$

B. $c = \frac{t}{35} + 23$

C. $c = 23t + 35$

D. $c = 35t + 23$

$$y = 12x - 20$$

$$y = 28$$

What is the solution x, y to the given system of equations?

- A. 4, 28
- B. 20, 28
- C. 28, 4
- D. 28, 20

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under 220 miles. Rhett drove at an average speed of 35 miles per hour mph, and Jessica drove at an average speed of 40 mph. Which of the following inequalities represents this situation, where r is the number of hours Rhett drove and j is the number of hours Jessica drove?

- A. $35r + 40j > 220$
- B. $35r + 40j < 220$
- C. $40r + 35j > 220$
- D. $40r + 35j < 220$

Q7

GRID-IN ALGEBRA

The y -intercept of the graph of $12x + 2y = 18$ in the xy -plane is $0, y$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Nasir bought 9 storage bins that were each the same price. He used a coupon for \$63 off the entire purchase. The cost for the entire purchase after using the coupon was \$27. What was the original price, in dollars, for 1 storage bin?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function f is defined by $f(x) = 3x - 8$. What is the value of $f7$?

A. 29

B. 13

C. -5

D. -29

A dance teacher ordered outfits for students for a dance recital. Outfits for boys cost \$26, and outfits for girls cost \$35. The dance teacher ordered a total of 28 outfits and spent \$881. If b represents the number of outfits the dance teacher ordered for boys and g represents the number of outfits the dance teacher ordered for girls, which of the following systems of equations can be solved to find b and g ?

A. $26b + 35g = 28$
 $b + g = 881$

B. $26b + 35g = 881$
 $b + g = 28$

C. $26g + 35b = 28$
 $b + g = 881$

D. $26g + 35b = 881$
 $b + g = 28$